Backtest Overfitting Statistics under Serial Dependence and Benchmark Choice

A Replication and Empirical Stress Test: research brief

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Question. The probabilistic and deflated Sharpe ratios (PSR, DSR) and the probability of backtest overfitting (PBO) are often used as hurdles: "DSR > 0.95", "PBO ≤ 0.05". How often do these hurdles pass strategies with no edge when returns are serially dependent, heavy tailed or correlated, and does a historical search winner survive tests that allow for both the search and the benchmark? Motivating fact: the best of 100 zero-skill strategies over five years of daily data shows an annualized Sharpe ratio of about 1.12, which looks like a t-statistic of 2.5.

What was done. A test-driven reimplementation of PSR, MinTRL, DSR, MinBTL and CSCV/PBO that reproduces numerical examples from the four source papers, with page-level provenance. Zero-edge Monte Carlo (2000 datasets per cell) under IID, AR(1), GARCH(1,1), Student-t and equicorrelated returns. A circular block bootstrap of daily Kenneth French 49-industry returns, 1971–2026, with every mean set to zero. A search over 93 moving-average rules on the Fama–French market factor (1972–2026, 5 bps per position change), judged against zero skill and against the market by Sharpe ratio and by alpha, with Bonferroni and Hansen's SPA test.

Main findings.

1. **PSR's false-pass rate on real data depends on the period.** With each industry centred, uncorrected $\mathrm{PSR}(0) > 0.95$ passes 8.35% [7.17, 9.65] of resampled 1971–1998 histories and 3.85% [3.05, 4.79] of 1999–2026 histories, against a nominal 5%. The first excess survives a Holm adjustment over all eight real-data tests; the second is suggestive (Holm $p = 0.096$). Post hoc controls point to dependence as the cause: with dependence removed the rates are 5.05% and 4.67%, and the predicted late-period rate stays below 5% across 200 resampled histories and split dates from 1993 to 2004. Mean lag-1 autocorrelation fell from $+0.157$ to -0.026.
2. **The verdict on a search winner depends on the benchmark.** The selected rule (15/225) has Sharpe 0.588 against the market's 0.458. Against zero skill its DSR is 0.998; DSR of active returns (rule minus market) is 0.030. This tests a different estimand from the difference of standalone Sharpes or alpha. Its Sharpe advantage, $+0.130$ (s.e. 0.099), is not significant (one-sided $p = 0.094$; SPA $p = 0.504$), and the test has power 0.37 against a difference of that size. Its market alpha is 3.16% a year ($t = 2.71$, beta 0.51): it fails Bonferroni over 93 rules, but a post hoc SPA test on the 93 alphas gives $p = 0.035$–0.065 across block lengths, so the baseline alpha evidence is borderline. One extra day of delay gives alpha SPA $p = 0.084$ (Monte Carlo s.e. 0.0062), failing the 5% test. PBO is 0.208, but it falls to 0.051 and the winner changes when the signal is computed from the excess-return index instead.
3. **The 2014 DSR under correlated searches.** It passes at most 0.70% of zero-skill winners except under strong cross-strategy correlation (2.60% at $\rho = 0.7$; 5.05% with the interpolated $\hat{N} = 30.7$), because the cross-trial variance misses the common component.

Confirmations of published results. Uncorrected PSR passes 11.85% [10.47, 13.35] of zero-Sharpe AR(1) series with $\phi = 0.3$, against 11.37% from $F = (1 + \phi)/(1 - \phi)$ (Lo 2002; Ledoit and Wolf 2008; López de Prado, Lipton and Zoonekynd 2026); a 7-lag Newey–West variance leaves 7.10%. The 2014 DSR passes 0.05% of IID zero-skill winners, in line with the conservatism López de Prado and Porcu (2026) establish for its location-only form. Mean PBO stays near one half in zero-edge designs, as symmetry leads one to expect.

Decision	Statistic	A pass does not establish
One pre-specified Sharpe above a hurdle	PSR	Robustness to serial dependence
Search winner better than best of N noise trials	DSR (2014)	A 5% error rate; skill against a benchmark
Rankings persist out of sample	PBO	A probability of future loss
Move capital from benchmark to strategy	none of these	Needs search-aware Sharpe and alpha tests with power

Status of evidence. Designs and most predictions were recorded before each run; the hypotheses as stated, the evidence rule, the sub-period design and its prediction, the robustness checks and the review-stage checks (including the alpha tests) were fixed after earlier results were known and are labelled post hoc. A first sub-period run was withdrawn after its failed check revealed a centring error, and a walk-forward cost error was corrected (Sharpe 0.467 to 0.466). An internal adversarial review changed the case-study conclusion for market-adjusted return from "no evidence" to "borderline".

Main limitations. One data vendor and one market; PSR tested only at $\mathrm{SR}^* = 0$; fixed bootstrap block lengths; idealized same-close execution on a research portfolio; HAC assumptions unverified; over 100 tests with Holm adjustment only within tables; PBO power not measured; the 2026 DSR variants not tested.